AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

ANNUAL EXAMINATIONS (THEORY) 2025

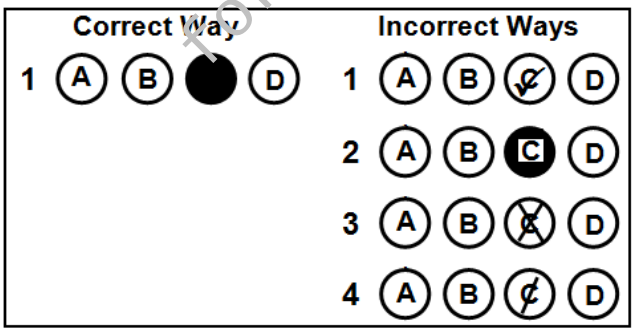
General Mathematics Paper I

Time: 1 hour 20 minutes Marks: 45

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 45 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. A formulae list is provided on page 2. You may refer to it during the paper, if you wish.
8. You may use a simple calculator if you wish.

List of Formulae

Note:

All symbols used in the formulae have their usual meaning.

Financial Mathematics

$$I = PT \times \frac{R}{100}$$

Basic Statistics

$$\overline{X} = \frac{\sum x}{n} \qquad \overline{X} = \frac{\sum fx}{n} \quad \text{or} \quad \overline{X} = \frac{\sum fx}{\sum f}$$

$$\text{Median} = l + \frac{1}{f}\left(\frac{n}{2} - c\right) \times h \qquad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{variance} = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 \qquad \text{Standard deviation} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

Quadratic Equations

$$ax^2 + bx + c = 0,\ a \neq 0 \qquad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \text{Disc} = b^2 - 4ac$$

Algebraic Manipulation

$$(a+b)^2 = a^2 + 2ab + b^2 \qquad (a-b)^2 = a^2 - 2ab + b^2 \qquad a^2 - b^2 = (a+b)(a-b)$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \qquad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Arithmetic and Geometric Sequence

$$a_n = a_1 + (n-1)d \qquad AM = \frac{a+b}{2} \qquad a_n = a_1 r^{n-1} \qquad GM = \pm\sqrt{ab}$$

Area and Volumes

$$\text{Area of a circle} = \pi r^2 \qquad \text{Area of a triangle} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{Volume of a cube} = l^3 \qquad \text{Volume of a cuboid} = l \times b \times h \qquad \text{Volume of a sphere} = \frac{4}{3} \times \pi r^3$$

$$\text{Volume of a cone} = \frac{1}{3} \times \pi r^2 \times h$$

$$\text{Volume of a cylinder} = \pi r^2 \times h$$

Introduction to Coordinate Geometry

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

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1. Shahab bought ten mobile phone sets from a Chinese exporter. Each mobile phone costs him 27,000 Pakistani Rupees.

   If one Chinese Yuan is equal to 29.15 Pakistani Rupees, then the amount he paid to the exporter, in Chinese Yuan, was

   A. 926.244
   B. 9,262.44
   C. 787,050
   D. 7,870,500

2. Samar earned a profit of Rs 45,000 at a simple interest rate of 6% per annum.

   If she invested an amount of Rs 100,000, then her investment period will be

   A. 2.7 years.
   B. 3.7 years.
   C. 7.5 years.
   D. 13.3 years.

3. Sana bought a refrigerator on installments. The details of the installment plan are given in the table.

Total Price (Rs)	98,000
Down Payment (Rs)	30% of the total price
Number of Monthly Installments	14
Processing Charges in Addition to the Total Price	7% of the total price

   The processing charges were

   A. Rs 2,058
   B. Rs 4,802
   C. Rs 6,860
   D. Rs 7,340

4. The sale price of a purse is Rs 2,200. If 10% of this amount is sales tax, then the sales tax levied on the purse is

   A. Rs 220
   B. Rs 1,980
   C. Rs 2,400
   D. Rs 2,420

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5. Tabinda works in a boutique. Her monthly salary is Rs 45,000. She also gets a bonus as per given schedule in the table.

Sales Target (Rs)	Bonus
75,000 to 100,000	5% of the total sale
100,001 to 200,000	7.5% of the total sale
200,001 to 300,000	8.5% of the total sale

In a certain month, she made a sale of Rs 180,000. The salary she gets will be

A. Rs 57,600
B. Rs 58,500
C. Rs 60,300
D. Rs 64,125

6. Alam works as a consultant for a corporate company. He works 15 hours a week and is paid Rs 500 each hour. He is also paid Rs 750 for each additional hour.

At this rate, if he works for 20 hours in a certain week, then his income for that week will be

A. Rs 7,500
B. Rs 10,000
C. Rs 11,250
D. Rs 15,000

7. For a certain telecommunication company, the given table shows call rates and taxes levied on call charges.

Call rate	CED rate	WH Tax	Additional
Rs 5 per call	10% per call	5% of total bill	0

(**Note**: CED stands for central excise duty and WH tax stands for withholding tax.)

If a household made 700 calls in the month of June, then telephone bill for the month will be

A. Rs 192.50
B. Rs 525.00
C. Rs 3,850.00
D. Rs 4,042.50

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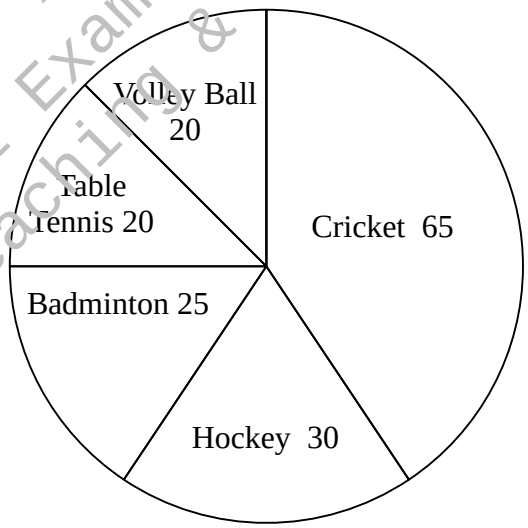
8. Given is the frequency distribution table of weights, in grams, of 50 apples.

Weight of Apples (in grams)	Number of Apples
30-34	4
35-39	8
40-44	16
45-49	12
50-54	10

The class mark of the class interval having the highest frequency is

A. 39.5
B. 41.0
C. 42.0
D. 44.5

9. A survey is conducted to obtain data about the favourite sports of students of class X. The result of the survey is represented by the given pie-chart.



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The percentage of the students who liked cricket is

A. 18.0%.
B. 40.6%.
C. 65.0%.
D. 68.4%.

10. The relationship between standard deviation s and variance σ^2 is

A. $s = \sigma^2$
B. $s^2 = \sigma$
C. $s = 2\sigma$
D. $2s = \sigma$

11. If a and b are positive integers and $a < b$, then the range of the data set $10 - a$, $10 + a$, 10, $10 - b$ and $10 + b$ will be

A. $2a$.
B. $2b$.
C. $a - b$.
D. $b - a$.

12. The median of the data set 10, 4, 6, 10, 4, 6 and 7 is

A. 6
B. 6.7
C. 7
D. 10

13. The runs scored by batters in a particular limited overs cricket tournament is presented in the given frequency polygon.



According to the given frequency polygon, the number of batters who scored between 60 and 105 runs is

A. 17
B. 40
C. 102
D. 119

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14. The highest common factor (HCF) of the algebraic expressions $(x-1)(x+2)$, (x^2-4) and $(x+2)^2$ is

A. $x+2$
B. $(x-1)(x+2)$
C. $(x-1)(x^2-4)$
D. $(x-1)(x-2)(x+2)^2$

15. The Least Common Multiple (LCM) of the expressions $(x-2)^2$ and x^2-4 is

A. $x-2$
B. $x+2$
C. $(x+2)(x-2)^2$
D. $(x-2)(x+2)^2$

16. The simplest form of the algebraic expression $\frac{2x^2-8y^2}{8} \div \frac{4}{(x+2y)(x-2y)}$ is

A. 1
B. $\frac{16}{(x+2y)^2(x-2y)^2}$
C. $(x+2y)^2(x-2y)^2$
D. $\frac{(x+2y)^2(x-2y)^2}{16}$

17. The simplified form of $\sqrt{\frac{64x^4y^2}{4x^2y^2}}$ is

A. $4x$
B. $4x^2$
C. $8x$
D. $8x^2$

18. The product of the least common multiple (LCM) and highest common factor (HCF) of two polynomials is $(x^3+8)(x+2)^2$.

If one of the polynomials is $(x+2)^2$, then the other polynomial will be

A. x^3+8
B. $(x+2)^2$
C. (x^2+2x+4)
D. (x^2+4x+4)

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19. For $\frac{a}{b}x = a$, the value of x is equal to

A. b.
B. $-b$.
C. $-\frac{a^2}{b}$.
D. $\frac{a^2}{b}$.

20. In the equation $\sqrt{x} - \sqrt{2} = \sqrt{2}$, the value of x will be

A. 0
B. 4
C. 8
D. 16

21. Sara bought 2 ice-creams and 2 chocolates for Rs 120.if each chocolate costs her Rs 20, then the price of each ice-cream will be

A. Rs 40
B. Rs 50
C. Rs 80
D. Rs 100

22. The solution set of $-\frac{1}{x} < -1$, where $x \in R$, will be

A. $\{x / x \in R \wedge x > 1\}$.
B. $\{x / x \in R \wedge x < 1\}$.
C. $\{x / x \in R \wedge x < -1\}$.
D. $\{x / x \in R \wedge x > -1\}$.

23. If 5 is added to twice of a number, the result is 35. The number is

A. 15
B. 20
C. 25
D. 30

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24. For the equation $\sqrt{x} = 2\sqrt{x-1}$, the value of x will be

A. $-\frac{4}{3}$

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. $\frac{4}{3}$

25. In the given equations, the equation(s) that can be reduced to quadratic equation(s) in standard form is/ are

I. $\frac{1}{x-3} = x$

II. $\frac{5x}{x+1} = x$

III. $x^2 - 2x + 5 = (1-x)^2$

A. I only.
B. III only.
C. I and II.
D. II and III.

26. The solution set of the quadratic equation $x^2 - x + 2 = 0$ is

A. $\left\{\frac{1 \pm \sqrt{-7}}{2}\right\}$

B. $\left\{\frac{1 \pm \sqrt{-7}}{4}\right\}$.

C. $\left\{\frac{-1 \pm \sqrt{-7}}{2}\right\}$.

D. $\left\{\frac{-1 \pm \sqrt{-7}}{4}\right\}$.

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27. The solution set of the quadratic equation $(ax+b)\times(bx-a)=0$ is

A. $\left\{-\frac{a}{b}, -\frac{b}{a}\right\}$.

B. $\left\{\frac{a}{b}, -\frac{b}{a}\right\}$.

C. $\left\{-\frac{a}{b}, \frac{b}{a}\right\}$.

D. $\left\{\frac{a}{b}, \frac{b}{a}\right\}$.

28. The length of a rectangle is double of its width. If its area is 32 cm^2, then its length would be

A. 16 cm.
B. 8 cm.
C. 4 cm.
D. 2 cm.

29. If the 1st term of a geometric sequence is 1 and the 3rd term is 9, then the 2nd term is

A. ± 3
B. ± 4
C. ± 6
D. ± 9

30. If the numbers −2, 4, *x*, 16 are in an arithmetic sequence, the value of *x* is

A. -8
B. 6
C. 8
D. 10

31. The 3rd term of a geometric progression is ax and the 4th term is x^3. The common ratio of this progression is

A. ax^2

B. $\frac{x^2}{a}$

C. $\frac{a}{x^2}$

D. $\frac{x}{a}$

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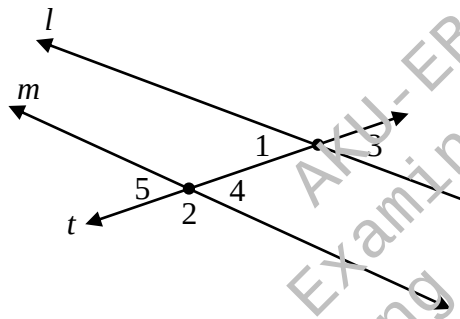
32. The geometric mean between $a-b$ and $a+b$ is

A. $\pm a$.
B. $\pm\sqrt{a^2+b^2}$.
C. $\pm\sqrt{a^2-b^2}$.
D. $\pm\dfrac{\sqrt{a^2+b^2}}{2}$.

33. If the arithmetic mean between two numbers a and 5 is 3, then the value of a is

A. -1
B. 1
C. 4
D. 11

34. In the given diagram, one of the pairs of vertical opposite angles is

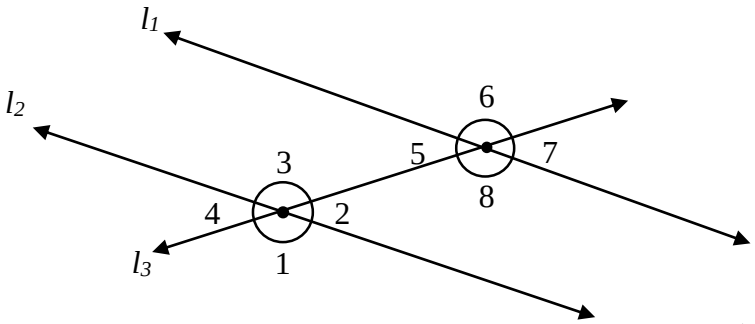


A. 5 and 2
B. 5 and 4
C. 1 and 4
D. 1 and 5

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Use the given information to answer Q.35 and Q.36.

In the given diagram, a transversal l_3 intersects the two parallel lines l_1 and l_2.



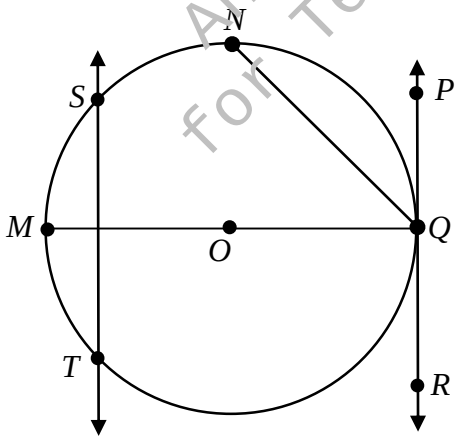
35. In the given diagram, one of the pairs of corresponding angles is

A. 3 and 4
B. 1 and 8
C. 4 and 6
D. 3 and 5

36. If $\angle 5 = x^o$, then $\angle 2$ will be equal to

A. $x°$
B. $(90 - x)°$
C. $(90 + x)°$
D. $(180 + x)°$

37. The given diagram shows a circle having centre at *O*.



In the given diagram, the tangent to the circle is

A. *ST*.
B. *PR*.
C. *MQ*.
D. *NQ*.

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38. If one edge of a cube is $\frac{l}{3}$ units, then the volume of cube, in cubic units, will be

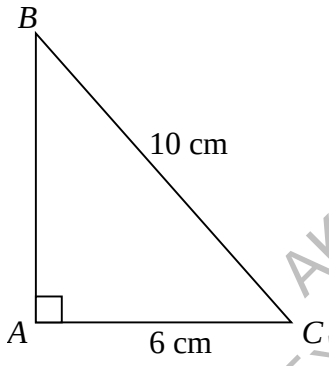
A. $\frac{l^3}{3}$

B. $\frac{l^3}{6}$

C. $\frac{l^3}{9}$

D. $\frac{l^3}{27}$

39. In the given triangle *ABC*, the length of *AB* is equal to



NOT TO SCALE

A. 4 cm.
B. 6 cm.
C. 8 cm.
D. 64 cm.

40. If the sides of an equilateral triangle *ABC* are of length 2 cm each, then the area of triangle *ABC* will be

(**Note**: $S = 3$ cm)

A. 6 cm^2
B. $2\sqrt{3}$ cm^2
C. 3 cm^2
D. $\sqrt{3}$ cm^2

41. If the volume of a hemisphere is a cubic units, then the volume of sphere will be

A. $\frac{a}{4}$ cubic units.

B. $\frac{a}{2}$ cubic units.

C. $2a$ cubic units.

D. $4a$ cubic units.

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42. A right circular cylinder has a height h and its base has a diameter d. The volume of the right circular cylinder is

A. $\pi d^2 h$.

B. $\pi \frac{d^2}{2} h$.

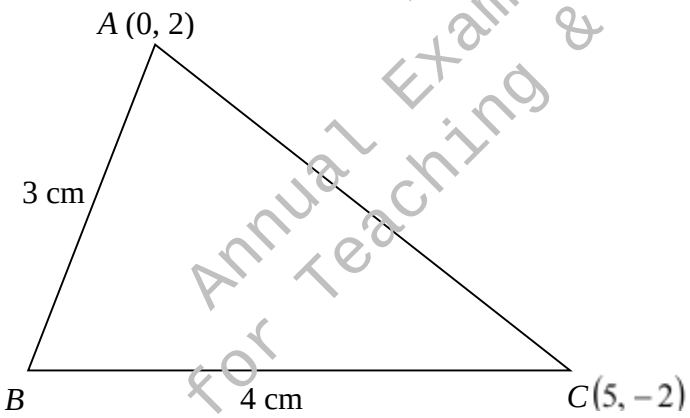
C. $\pi \frac{d^2}{4} h$.

D. $\pi \left(\frac{d}{4}\right)^2 h$.

43. The length of the line segment joining $(2, 0)$ and $(0, 2)$ is

A. 2 units.
B. 4 units.
C. $2\sqrt{2}$ units.
D. $4\sqrt{2}$ units.

44. If the coordinates of the vertices A and C are $(0, 2)$ and $(5, -2)$ respectively, then the given triangle is



NOT TO SCALE

A. a right angled triangle.
B. an equilateral triangle.
C. an isosceles triangle.
D. a scalene triangle.

45. If the midpoint of (2, 0) and (a, 2) is (3, 1), then the value of a is

A. 1
B. 2
C. 4
D. 6

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END OF PAPER

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